

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 12A.2

TOTAL ENTROPY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	12: Entropy and Energetics
SUB-TOPIC:	12A.2: Total Entropy
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	4 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 12A.2

Q1. For a reaction, $\Delta H = -92.2 \text{ kJ mol}^{-1}$. Calculate the entropy change of the surroundings at 298 K.

[WCH14/01/Jan23/Q5(b)]

[2 Marks] WCH14/01

Q2. The entropy change of the system is $-198.8 \text{ J K}^{-1} \text{ mol}^{-1}$.

Use the answer from the previous part to calculate the total entropy change at 298 K. [WCH14/01/Jan23/Q5(c)]

[2 Marks] WCH14/01

SECTION B: WORKED MARK SCHEME — 12A.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARK S
Q1	Q1: For a reaction, $\Delta H = -92.2 \text{ kJ mol}^{-1}$. Calculate the entropy change of the surroundings at 298 K. [WCH14/01/Jan23/Q5(b)] $\Delta S_{\text{surroundings}} = -\Delta H / T = -(-92200 \text{ J mol}^{-1}) / 298 \text{ K (1)} = +309.4 \text{ J K}^{-1} \text{ mol}^{-1} \text{ (1)}$	[2]
Q2	Q2: The entropy change of the system is $-198.8 \text{ J K}^{-1} \text{ mol}^{-1}$. Use the answer from the previous part to calculate the total entropy change at 298 K. [WCH14/01/Jan23/Q5(c)] $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} = -198.8 + 309.4 \text{ (1)} = +110.6 \text{ J K}^{-1} \text{ mol}^{-1} \text{ (1)}$	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 12A.2**FAQ #1: Why do students lose marks in calculating Delta S_{surroundings}?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why do students lose marks in calculating Delta S_{surroundings}?

MODEL ANSWER / EXAMINER EXPECTATION:

Candidates forget to convert Delta H from kJ mol⁻¹ to J mol⁻¹ before dividing by temperature in Kelvin. This results in a value that is 1000 times too small.

FAQ #2: What is the common sign error for Delta S_{surroundings}?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is the common sign error for Delta S_{surroundings}?

MODEL ANSWER / EXAMINER EXPECTATION:

Students often forget the negative sign in the equation $\Delta S_{\text{surr}} = -\Delta H / T$. An exothermic reaction ($-\Delta H$) always leads to a positive Delta S_{surroundings}.